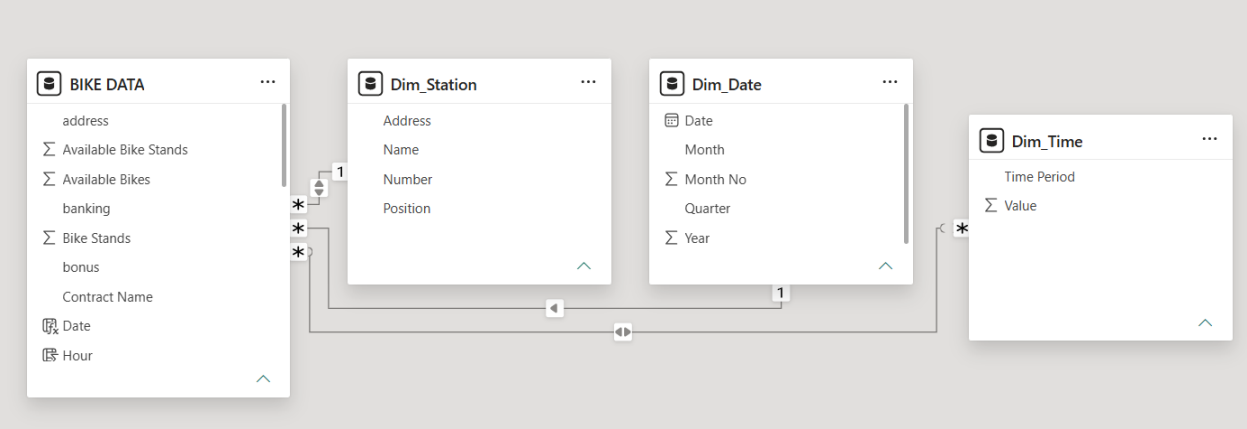
Analyzing Smart City Bike Sharing Data using Power BI

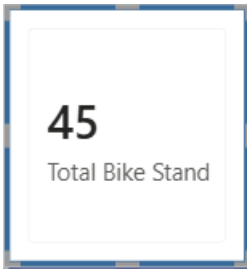
OBJECTIVES

The objective of this project is to analyze the performance of the bike-sharing system using Power BI. It helps monitor bike availability, station status, and bike usage across different locations. The dashboard identifies high-demand stations and peak usage periods through monthly and hourly trend analysis. It provides interactive visualizations and KPIs to support data-driven decision-making. The insights from this project help optimize bike distribution, improve operational efficiency, and enhance the overall user experience.

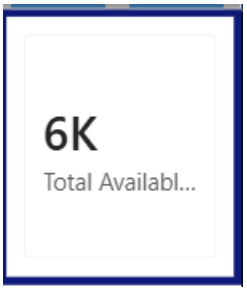
Data Modeling



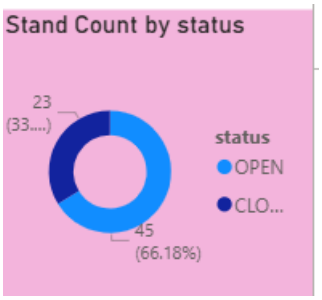
1.How many bike stations are available?



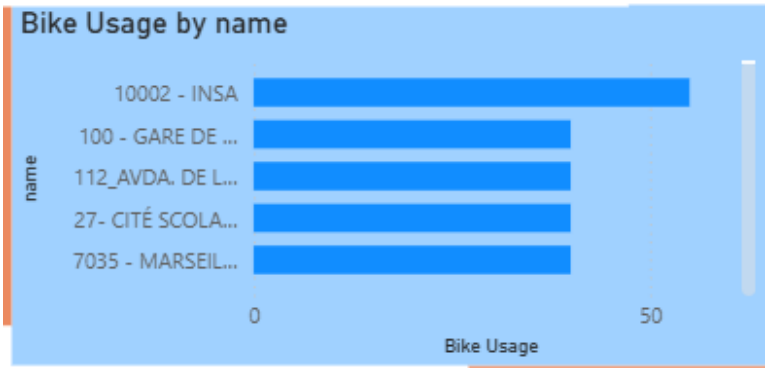
2.What is the total number of bikes available?



3.How many stations are OPEN and CLOSED?



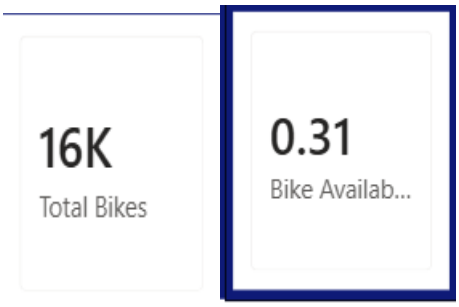
4.Which station has the highest bike usage?



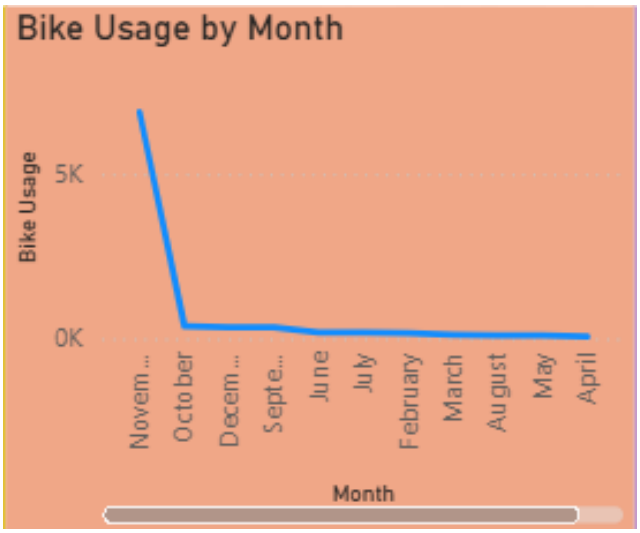
5.Which locations have the highest bike demand?



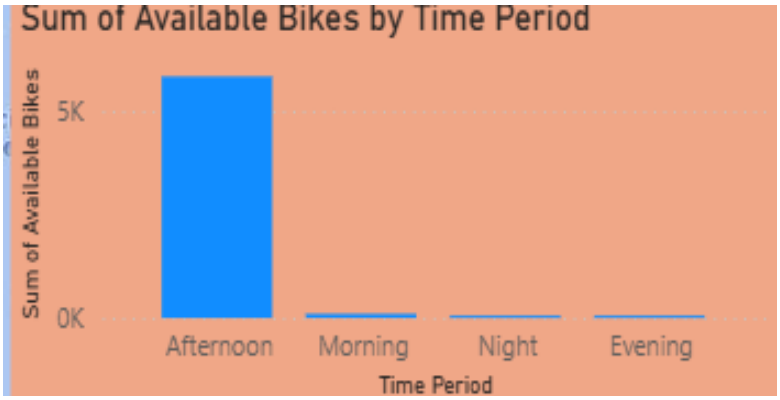
6.What is the total bike & bike availability?



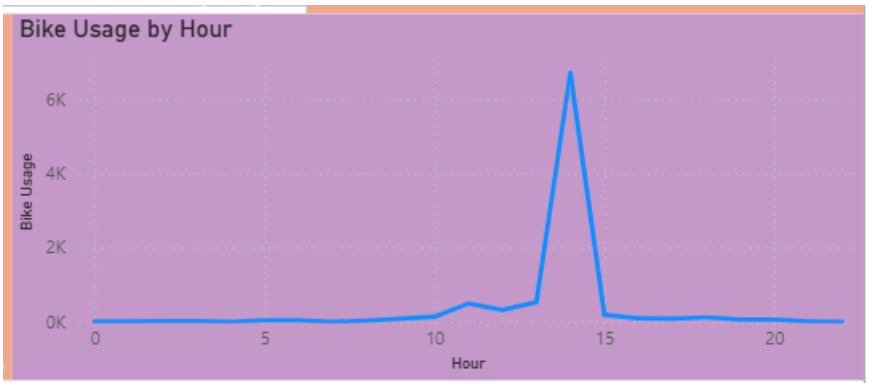
7. Which month records the highest bike usage?



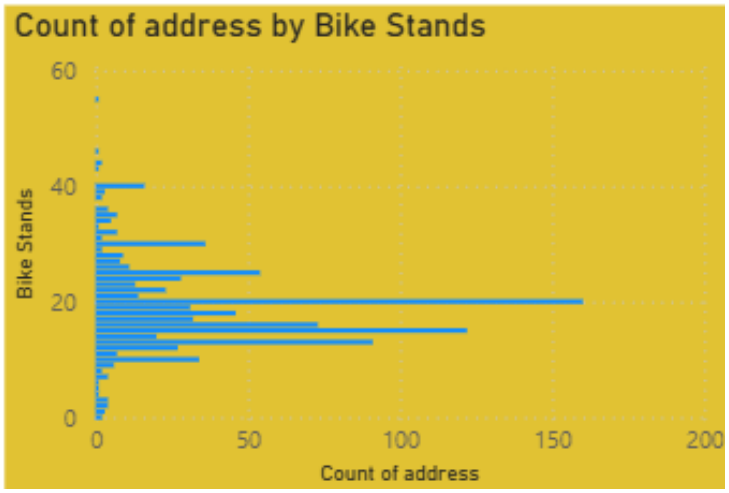
8.During which time period is bike usage the highest?



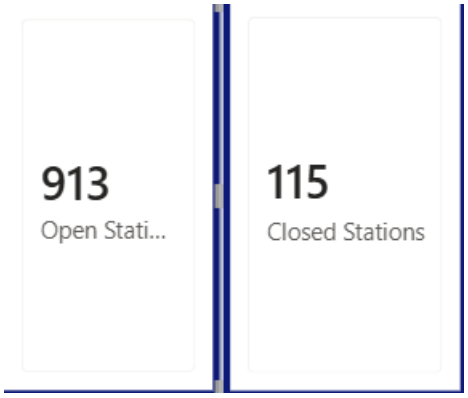
9. During which time period are the most bikes available?



10. Which addresses have the highest number of bike stations?



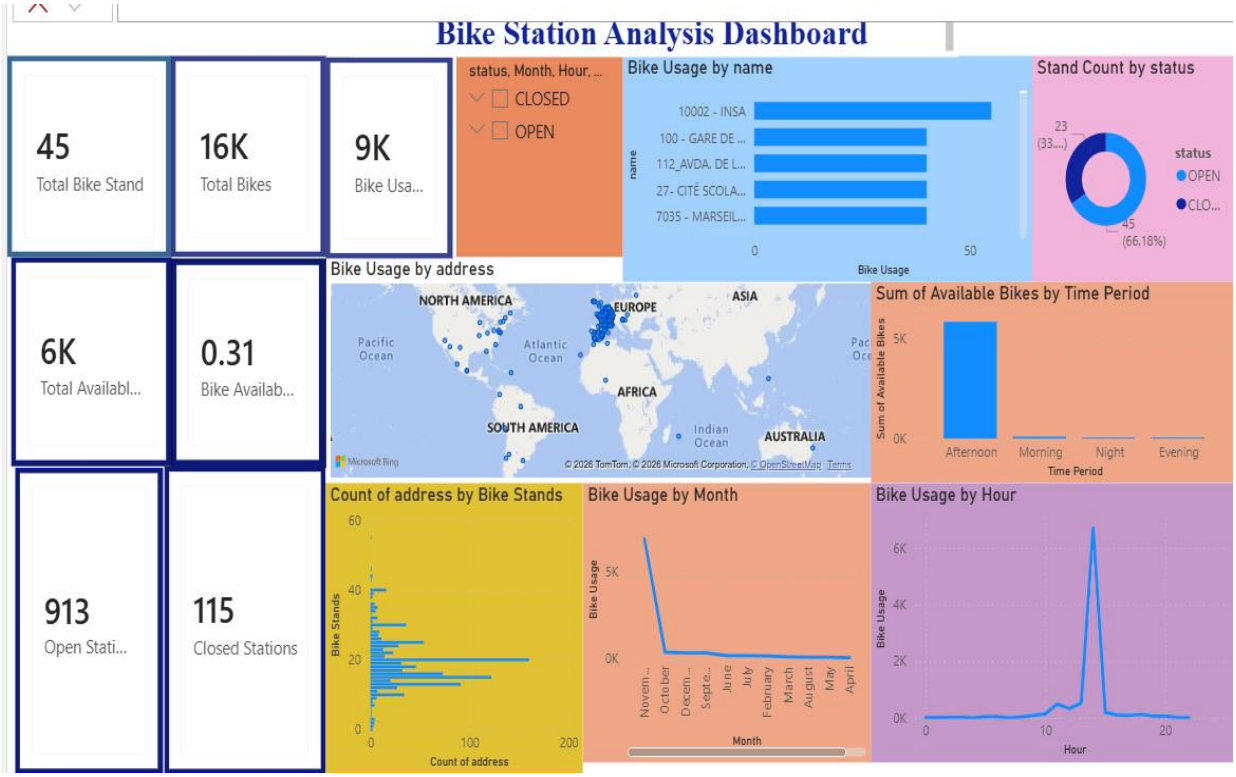
11. How many stations are OPEN and CLOSED?



DAX Measures

1. Total Bike Stands = SUM ('BIKE DATA'[Bike Stands])
2. Total Available Bikes = SUM ('BIKE DATA'[Available Bikes])
3. Bike Usage = SUM ('BIKE DATA'[Bike Stands]) - SUM('BIKE DATA'[Available Bikes])
4. Bike Availability % = DIVIDE ([Total Available Bikes], [Total Bike Stands], 0) * 100
5. Open Stations = CALCULATE (COUNTROWS ('BIKE DATA'), 'BIKE DATA'[Status] = "OPEN”)
6. Closed Stations = CALCULATE (COUNTROWS ('BIKE DATA'), 'BIKE DATA'[Status] = "CLOSED”)
7. Total Stations = DISTINCTCOUNT ('BIKE DATA'[Number])
8. Average Available Bikes = AVERAGE ('BIKE DATA'[Available Bikes])
9. Zero Available Bike Stations = CALCULATE (COUNTROWS ('BIKE DATA'), 'BIKE DATA'[Available Bikes] = 0)
10. Average Bike Usage = DIVIDE ([Bike Usage], [Total Stations], 0)

Data Visualization & Dashboard



Conclusion & Insights

The Bike Sharing Analysis Dashboard provides valuable insights into station performance and bike availability. The findings can help operators optimize bike distribution, reduce shortages at high-demand stations, and improve service efficiency. This dashboard supports data-driven decision-making, leading to better resource management and an enhanced user experience.